



# FEBAUTO Racing

## Competitor manual

- Part I** The platform and the rules
- Part II** From nothing to a leaderboard entry
- Part III** Learning path and reference

Formula Electric at Berkeley  
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<https://github.com/Pranman1/feb-racing>  
<https://pranman1.github.io/feb-racing/>

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## Part I

# The platform

## 1 What it is

FEBAUTO Racing is a sim-racing programme for autonomous driving. Members write a ROS 2 driving stack for a 1:10 scale RoboRacer car, run it in the FEB Simulator on their own laptop, and post lap times to a public leaderboard. Twice a semester the organiser runs a competition on a secret track, with the same rules as the RoboRacer Sim Racing League.

The simulator is a fork of the AutoDRIVE Simulator (Unity). The car, its sensors and its physics are the league's RoboRacer model, unchanged, so a stack that works here works in the league too.

### 1.1 The pieces

| Piece                                                                              | What it does                                                                                                                                                                 | Who touches it                             |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| FEB Simulator                                                                      | Native app (Mac, Windows, Linux). Simulates the car on a track folder and talks to a bridge over a WebSocket on port 4567.                                                   | Everyone runs it, nobody edits it          |
| Devkit container                                                                   | Docker image <code>ghcr.io/pranman1/feb-devkit</code> : ROS 2 Humble plus the AutoDRIVE bridge that turns the WebSocket into ROS 2 topics. The member's code runs inside it. | Members' code runs inside it               |
| <code>stack/</code>                                                                | The member's ROS 2 packages. Mounted into the container and built on every start.                                                                                            | Members                                    |
| <code>tracks/</code>                                                               | Public tracks. PNG and YAML in, <code>track.json</code> out. Shipped inside the app.                                                                                         | Organiser adds, members use                |
| <code>feb-sim</code>                                                               | The member launcher: <code>setup</code> , <code>run</code> , <code>practice</code> , <code>submit</code> , <code>shell</code> , <code>stop</code> .                          | Members                                    |
| <code>feb-race</code>                                                              | The organiser harness: events, nightly reruns, verification, head-to-head brackets.                                                                                          | Organiser                                  |
| <code>results/</code> ,<br><code>events/</code> ,<br><code>submissions.yaml</code> | The data behind the leaderboard site.                                                                                                                                        | Harness writes, members' pull requests add |
| Site                                                                               | GitHub Pages, rebuilt on every push to <code>main</code> .                                                                                                                   | Read-only                                  |

### 1.2 Data flow while driving

```
lidar, imu, encoders, camera, lap/collision counts --> bridge --> /autodrive/roboracer_1
/* --> your node
your node --> /autodrive/roboracer_1/steering_command, throttle_command --> bridge -->
simulator
```

The car takes exactly two inputs and everything else is a sensor:

- `steering_command`, `std_msgs/Float32` in  $[-1, 1]$ : full left to full right,  $\pm 0.5236$  rad. The actuator lags about 0.25 s and slews at 3.2 rad/s.
- `throttle_command`, `std_msgs/Float32` in  $[-1, 1]$ . **Zero is a hard brake**, not coasting. A car that wants to keep moving must never send zero.

### 1.3 Sensors (namespace /autodrive/roboracer\_1/)

| Topic                                                                    | Type                   | Notes                                              |
|--------------------------------------------------------------------------|------------------------|----------------------------------------------------|
| lidar                                                                    | sensor_msgs/LaserScan  | 1080 beams over 270°, 0.06 to 10 m, 40 Hz          |
| imu                                                                      | sensor_msgs/Imu        | orientation, angular velocity, linear acceleration |
| left_encoder,<br>right_encoder                                           | sensor_msgs/JointState | wheel angle in rad; wheel radius 0.059 m           |
| steering, throttle                                                       | std_msgs/Float32       | actuator feedback                                  |
| front_camera                                                             | sensor_msgs/Image      | forward camera (cone tracks)                       |
| ips, odom, tf                                                            | pose                   | ground truth: <b>training and debugging only</b>   |
| lap_count, lap_time,<br>last_lap_time, best_lap_time,<br>collision_count | counters               | scoring: <b>training and debugging only</b>        |
| reset_command                                                            | command                | respawn: <b>training and debugging only</b>        |

Wheelbase 0.324 m, half width 0.135 m, steady-state speed about 23 m/s per unit throttle. With a window open the bridge exchanges data at about 10 Hz (each exchange reads the cameras back from the GPU); headless it reaches 20 to 40 Hz. Scored runs go through a proxy fixed at 10 Hz for everyone, so a driver must cope with 10 Hz. The starter kit does.

## 2 Rules

These follow the RoboRacer Sim Racing League.

- One attempt is 1 warm-up lap plus 10 timed laps. Total time is the sum of the timed laps plus **10 s per collision**.
- Not finishing within **300 s** is DNF, ranked after every finisher by laps completed. More than **10 collisions** is DSQ.
- Each team gets the number of attempts stated in the event; the best counts.
- **Qualifying for the first competition:** on *every* practice track, a verified practice run of 10 laps with at most 1 collision. Practice tracks are added through the semester; the bar applies to all of them. The Teams page of the site shows each team's progress per track.
- **The final** is head-to-head, two cars at a time, in a knockout seeded by the qualification standings; only the top seeds (set per event, 8 by default) enter.
- **Restricted topics at race time:** `ips`, `odom`, `tf`, the lap and collision counters, and `reset_command`. The harness records which nodes subscribe to them during the run and flags the result on the site. Use them freely while developing.
- Everything a submission needs starts from the container's entrypoint (`/home/autodrive_devkit.sh`), never from `.bashrc`.
- Head-to-head: car-to-car contact counts as a collision for both cars and respawns them side by side. Finishing order is most laps first, then the finish time including penalties.

## Part II

# From nothing to a leaderboard entry

### 3 Install (once, about 15 minutes)

#### 3.1 Prerequisites

| OS      | Install                                                                                                                                                                                                                                                                                                           |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| macOS   | Docker Desktop ( <a href="https://docker.com">https://docker.com</a> ), Python 3 (comes with the Xcode command line tools: <code>xcode-select -install</code> ), GitHub CLI ( <code>brew install gh</code> ), Foxglove (the .dmg from <a href="https://foxglove.dev/download">https://foxglove.dev/download</a> ) |
| Windows | Docker Desktop with the WSL 2 backend, Python 3 from python.org, GitHub CLI ( <code>winget install GitHub.cli</code> ), Foxglove (the installer from <a href="https://foxglove.dev/download">https://foxglove.dev/download</a> ). Run the commands below in Git Bash or WSL.                                      |
| Linux   | <code>sudo apt install docker.io python3 gh</code> , then add yourself to the <code>docker</code> group and log in again; Foxglove: <code>sudo apt install ./foxglove-studio-*.deb</code> from <a href="https://foxglove.dev/download">https://foxglove.dev/download</a>                                          |

Start Docker Desktop before anything else. Log the GitHub CLI in once: `gh auth login`. Open Foxglove once and sign in (free account); it is how you look at your car's data, see section 4.4.

#### 3.2 Get the repo and the simulator

```
git clone https://github.com/Pranman1/feb-racing
cd feb-racing
./feb-sim setup --team "Your Team"
```

`setup` checks Docker, pulls the devkit image, downloads the simulator app for your OS into `~/feb-sim/app/`, and checks that Foxglove is installed (it opens the download page if not). The team name is what your results are posted under; use the name your team registers in `submissions.yaml`.

#### 3.3 First drive: manual

```
./feb-sim run
```

A window opens on the *FEB Loop* track in Driver's Eye view. Nothing drives by itself: with no stack given, the launcher starts no driver and no opponent. Arrow keys steer and throttle. The menu in the top-left has the Manual/Autonomous button, the Track button (cycles the built-in tracks), the Cars button (more cars on the grid), the Look button (Visual or Simple scene; Simple for a weak laptop) and the camera choices. The HUD at the bottom shows the running lap time (its clock starts when you first cross the start line), last and best lap, lap count, collisions and speed. After your first completed lap a translucent gold ghost replays your best lap so you can see where you gain and lose time.

**macOS, if the app does not open** (“damaged” or “cannot be opened”): the app is not signed by Apple. Run once:

```
xattr -dr com.apple.quarantine ~/feb-sim/app/mac/"FEB Simulator.app"
codesign --force --deep -s - ~/feb-sim/app/mac/"FEB Simulator.app"
```

## 4 Write your driver

### 4.1 Where the code lives

Copy the starter kit into your own folder. `stack/` is yours and is ignored by git in this repo, so nothing you do there conflicts with updates.

```
cp -r starter_kit/feb_driver stack/
```

|                                            |                                                                           |
|--------------------------------------------|---------------------------------------------------------------------------|
| <code>stack/feb_driver/</code>             |                                                                           |
| <code>feb_driver/reactive_driver.py</code> | THE DRIVER. Subscribes to lidar + encoders, publishes steering + throttle |
| <code>config/driver.yaml</code>            | its tuning knobs (speeds, gap bubble, filters, throttle law)              |
| <code>launch/drive.launch.py</code>        | starts the driver with the yaml                                           |
| <code>feb_driver/sysid_probe.py</code>     | stage 2: drives test inputs and records a bag                             |
| <code>tools/sysid_fit.py</code>            | stage 2: fits the recorded bag                                            |
| <code>package.xml, setup.py</code>         | standard ROS 2 Python package files                                       |

`reactive_driver.py` is about 170 lines. Per lidar scan it cleans the scan, zeroes a safety bubble around the nearest return, aims into the widest gap, picks a speed from the room ahead and the steering angle, and turns that into a throttle as feed-forward plus a bounded trim. Everything is commented. `driver.yaml` exposes every number.

You can add more packages next to `feb_driver`, more nodes, C++ packages, anything that builds with `colcon`. The container builds everything under `stack/` on start.

### 4.2 Run it on the simulator

```
./feb-sim run --stack "ros2 launch feb_driver drive.launch.py"
```

What happens, in order: the launcher starts the devkit container with `stack/` mounted inside it; the container builds your packages and starts the bridge; the simulator starts, connects to the bridge, and is put in Autonomous mode; the container starts your launch command. The car drives.

Edit, save, run the same command again. Each run is a fresh build (a few seconds for Python).

### 4.3 Useful options

| Option                        | Effect                                                                                                                                                                                        |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-track porto</code>     | another built-in track ( <code>loop</code> , <code>loop_cones</code> , <code>porto</code> ), or a path to any track folder                                                                    |
| <code>-mode manual</code>     | keep the keyboard even with a stack running                                                                                                                                                   |
| <code>-opponent normal</code> | the house racer in car two ( <code>slow</code> 1.5, <code>normal</code> 2.0, <code>fast</code> 2.5 m/s, or <code>map</code> ). Needs <code>-stack</code> . Not on Porto (single-car track).   |
| <code>-cars 2</code>          | a second, undriven car on the grid                                                                                                                                                            |
| <code>-noise 1</code>         | lidar range noise and dropouts; <code>-noise "range_sigma:=0.05 latency:=0.05"</code> sets them                                                                                               |
| <code>-lidar-hz 10</code>     | slower lidar, as a real bridge often delivers                                                                                                                                                 |
| <code>-camera Trackcam</code> | start in another view ("God's Eye", <code>Trackcam</code> , "Driver's Eye")                                                                                                                   |
| <code>-no-foxglove</code>     | do not open Foxglove on the car's topics                                                                                                                                                      |
| <code>-look simple</code>     | the bare scene instead of the dressed one ( <i>Visual</i> : asphalt, banded walls, start line, sky, livery). Purely cosmetic; the menu's Look button toggles it and the choice is remembered. |

Car one is always *your* car: the Manual/Autonomous button, the keyboard, the Driver's Eye camera and topic namespace `roboracer_1` all belong to it. Your container never sees any other car.

#### 4.4 See the data: Foxglove

The devkit runs a Foxglove bridge on port 8765, so the Foxglove app on your laptop shows every topic live on any OS, with no ROS installed outside the container. With the app installed and signed in (prerequisites above), `./feb-sim run` opens Foxglove connected to your car, a few seconds after the simulator window. Without the app installed the launcher prints the address instead: `ws://localhost:8765` under *Open connection, Foxglove WebSocket*.

Useful panels: 3D (lidar as `/autodrive/roboracer_1/lidar`), Image (the front camera), Plot (`throttle_command`, `steering_command` and the encoders against time), Raw Messages and the Topics list with rates. Foxglove also opens the bags that `practice` records under `runs/<id>/`. `-no-foxglove` skips the auto-open.

#### 4.5 Debug

```
./feb-sim shell          # a ROS 2 shell inside the running container
ros2 node list
ros2 topic hz /autodrive/roboracer_1/lidar
ros2 topic echo /autodrive/roboracer_1/throttle_command
rqt
./feb-sim stop          # stop the container(s) and the simulator
```

The container log (build errors, your node's output) is `docker logs feb-devkit`.

#### 4.6 Keeping up to date

```
./feb-sim update
```

One command pulls the repo (new practice tracks, launcher and docs), pulls the newer devkit image if there is one, and replaces the simulator app if the organiser has released a new build. New tracks need no app update: the launcher hands the repo's `tracks/` folder to the app, so a pulled track appears in the Track button right away. `./feb-sim run` prints one line when a newer app exists.

### 5 Score yourself and post it

```
./feb-sim practice --stack "ros2 launch feb_driver drive.launch.py" --track loop
./feb-sim submit
```

`practice` runs one attempt under competition rules: warm-up plus 10 timed laps, 10 s per collision, DNF after 300 s, through the 10 Hz proxy. It prints the result and saves `runs/<id>/` with `result.json`, logs and a ROS bag.

`submit` opens a pull request with the result file. A check validates it and merges it, and you appear on that track's practice board as *unverified*. The organiser's nightly rerun of your image turns it *verified*.

### 6 Race the house racer

```
./feb-sim run --stack "ros2 launch feb_driver drive.launch.py" --opponent normal
```

The house racer starts in car two only when your stack starts your car. It is the smooth lidar driver at a conservative pace with zero wall contact: the thing a new stack has to overtake. It follows a car ahead rather than ramming it. `-opponent map` runs the map-following driver instead (it knows the track and the car's true pose; faster, less smooth). On cone-only tracks the launcher always uses the map follower, since a gap follower has no walls to follow there.

## 7 Enter competitions

The season has three steps. First, **qualify**: complete 10 laps with at most one collision on every practice track, verified by the organiser's nightly rerun of your image (so the image has to be registered, below). The Teams page of the site shows a tick per track and a *qualified* badge when all are done. Second, the **qualification competition**: a time-attack on a secret track, open to qualified teams. Third, the **final**: a head-to-head knockout, two cars per race, for the top seeds of the qualification.

Competitions run on the organiser's PC from a Docker image you publish. Build it from the starter Dockerfile, which starts from the devkit image, copies `stack/` in, builds it and sets the launch command:

```
docker build -t <dockerhub-user>/feb-racer:v1 -f starter_kit/Dockerfile stack/
docker push <dockerhub-user>/feb-racer:v1
```

If your launch command differs from the starter's, change the `FEB_LAUNCH` line in a copy of the Dockerfile. Then register: add your team, GitHub logins and image to `submissions.yaml` in a pull request:

```
teams:
- name: Your Team
  members: [github-login-1, github-login-2]
  image: docker.io/<dockerhub-user>/feb-racer:v1
```

Competitions and nightly reruns pull that image. Update the tag and the yaml whenever you want a new version raced. In a head-to-head final your container runs unchanged: a proxy between the simulator and the containers makes every car look like `roboracer_1` to its own stack.

## Part III

# Learning path and reference

## 8 Learning path

Every stage ends with a number on the leaderboard.

1. **Read the car.** `./feb-sim shell`, then `ros2 topic hz` and `ros2 topic echo` on the topics in Part I. Steering lags about 0.25 s; throttle 0 brakes hard.
2. **Reactive driving.** Tune `driver.yaml`: bubble, gap field of view, the deep-beam weight, speed limits, braking margin, filter time constants. Target: 10 clean laps on `loop` under 100 s.
3. **System identification.** `ros2 launch feb_driver sysid.launch.py` drives throttle steps, a coast and a chirp while recording a bag into `stack/sysid_bag`. Inside the container, `python3 tools/sysid_fit.py stack/sysid_bag` gives the steady-state gain (m/s per throttle), the idle-brake deceleration and the throttle and steering lags. Put the gain into `speed_per_throttle`. Expect the tyres to be a knife edge: lateral grip peaks at 1 % slip and halves by 10 %.
4. **Localisation and a map.** Build a map of the track from lidar (scan matching, or SLAM Toolbox in the container), then localise on it at race time. Check against `ips` while developing; never subscribe to it at race time.
5. **Raceline and controller.** A minimum-curvature line on your map with a speed profile from your sysid numbers, tracked by pure pursuit, then MAP or MPC. Lap times halve here.
6. **Realism.** `-noise`, `-lidar-hz 10`, `-opponent`, and the cone track (`loop_cones`: blue cones left, yellow right, a camera perception project).

## 9 If something goes wrong

| Symptom                 | Cause and fix                                                                                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Nothing drives          | Correct without <code>-stack</code> . With a stack: the build failed or the node crashed, see <code>docker logs feb-devkit</code> ; or the mode button says Manual. |
| Port 4567 in use        | Another bridge container is running: <code>./feb-sim stop</code> .                                                                                                  |
| Car surges and stops    | Throttle reaching zero (a hard brake). Keep feed-forward plus a bounded trim; see the starter's throttle law.                                                       |
| Car weaves at full lock | Raw beam-to-beam targets at 10 Hz. Low-pass and rate-limit steering; the starter shows how.                                                                         |
| Docker pull denied      | Docker Desktop not running, or the image is private; ask the organiser.                                                                                             |
| Mac app will not open   | See the <code>xattr</code> and <code>codesign</code> commands in the install section.                                                                               |
| Cones or walls missing  | The track folder has no <code>track.json</code> ; run <code>tools/track_build.py &lt;folder&gt;</code> .                                                            |

## 10 Command reference

| Command                                                                                                              | What it does                                               |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <code>feb-sim setup -team T</code>                                                                                   | pull the devkit image, download the app, set the team name |
| <code>feb-sim run [-track] [-stack] [-mode] [-opponent] [-cars] [-noise] [-lidar-hz] [-camera] [-no-foxglove]</code> | start the container, the simulator and Foxglove, connected |
| <code>feb-sim practice -stack S [-track] [-laps]</code>                                                              | one scored attempt into <code>runs/&lt;id&gt;/</code>      |
| <code>feb-sim submit [-run DIR]</code>                                                                               | pull request with the newest (or given) result             |
| <code>feb-sim update</code>                                                                                          | git pull, newer devkit image, newer simulator app          |
| <code>feb-sim shell</code>                                                                                           | ROS 2 shell inside the running container                   |
| <code>feb-sim stop</code>                                                                                            | stop containers and the simulator                          |

## 11 Repository layout

```
feb-racing/  
  feb-sim           the launcher (Python 3, needs only docker and gh)  
  starter_kit/      feb_driver package + submission Dockerfile  
  stack/            YOUR code (gitignored)  
  tracks/           public tracks (also shipped inside the app)  
  runs/             your attempt folders (gitignored)  
  results/, events/, submissions.yaml, site/  leaderboard data and page  
  docs/             handbook, this manual, detail pages
```